

App Usage Time by Age

(do NOT include your name)

2024/10/28

1 Abstract

In this report, the relationship between age and mobile app usage is studied to find a potential correlation.

The data, covering 700 mobile phone users aged between 18 and 60, could not show any correlation between age and time spent on using apps.

2 FSA

3 Introduction

With the increased prevalence of mobile phones in modern society, the health impacts of screen time and phone use has become subject to much discussion. It is often also claimed that younger generations spend too much time on their phones, often with negative sentiments. In light of this, the research question of this study reads as follows:

”Is there a relation between age and phone app usage?”

4 Related work

The correlation between age and screen time has previously been researched. For example, in Andone et al. (2016), where, by studying gender and age in relation to mobile phone usage, it is found that teenagers have the highest mean daily phone usage, a number that then decreases with the increase of age.

5 Data and Method

The data used for this study consists of a simulated dataset in csv-format covering different types of mobile phone usage of different demographics Khorasani (2024)¹. The code creates a scatterplot with regression line and automatically calculated the required values.

¹<https://www.kaggle.com/datasets/valakhorasani/mobile-device-usage-and-user-behavior-dataset/data>

6 Results

As can be seen in Figure 1, no correlation between age and app usage could be found in the examined dataset.

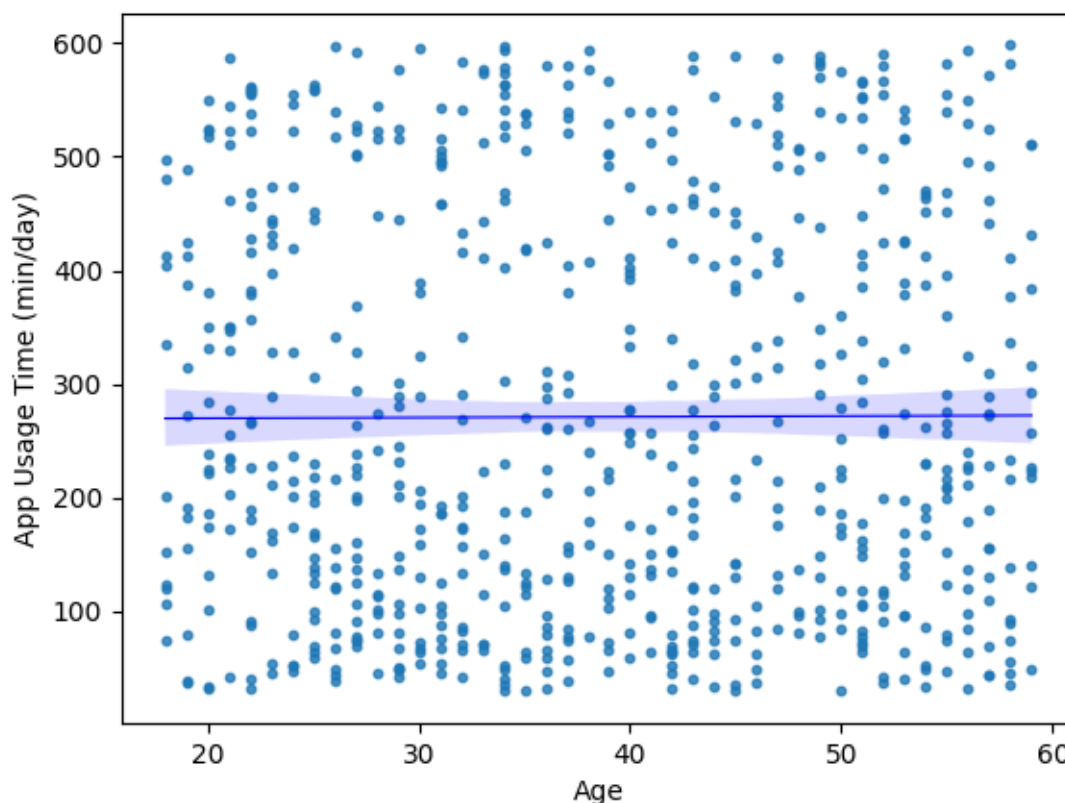


Figure 1: The relation between age (x-axis) and mobile phone app usage (min/day) (y-axis).

Based on the data, the following are the calculated values:

- Intercept: 268.64099168973155 (The value of y when x is 0)
- Coefficient: 0.06464124349201568 (Very low increase of y per increase of x)
- R-squared: 1.920398564167966e-05 (Model explains almost none of the data-variability - no relationship x and y)
- P-value: 0.90786146467596 (This high value means that there is almost no significance to any found relationship)

7 Concluding Discussion

As seen in 6, results of this study show no relationship between age and app usage. This is, however, likely due to the dataset not consisting of real data, but rather simulated data with the purpose of being as diverse as possible, rather than as based on reality as possible.

References

- Andone, I., Błaszkiwicz, K., Eibes, M., Trendafilov, B., Montag, C. and Markowetz, A. (2016), ‘How age and gender affect smartphone usage’.
- Khorasani, V. (2024), ‘Mobile device usage and user behavior dataset’, <https://www.kaggle.com/datasets/valakhorasani/mobile-device-usage-and-user-behavior-dataset/data>. Accessed: 2024-10-28.